Safety and Rollback Mechanisms for Objective Misinterpretation in Agentic AI Systems

Abstract—**Agentic artificial intelligence (AI) systems increasingly combine large language models (LLMs), planning, memory, tool use, code execution, and autonomous multi-step decision-making. These capabilities create a distinctive safety problem: an agent may retain substantial competence while pursuing an objective that differs from the designer's or user's intended objective. This phenomenon encompasses specification gaming, reward hacking, goal misgeneralization, reward tampering, and forms of inner misalignment. Conventional safeguards—reward-model improvement, human feedback, monitoring, constrained action spaces, and shutdown mechanisms—primarily seek to prevent or detect undesirable behavior before or during execution. This paper examines a complementary safety paradigm: rollback, defined as the controlled restoration of an agent and its execution environment to a previously validated state when objective misinterpretation is suspected. Drawing on research in AI alignment, reinforcement learning, corrigibility, safe reinforcement learning, checkpoint/restore systems, and recent agent-sandbox research, the paper develops a layered framework in which objective uncertainty, runtime monitoring, action gating, checkpointing, rollback, and post-rollback revalidation operate as mutually reinforcing controls. The analysis distinguishes logical rollback of agent state from physical or external-world reversibility and argues that the latter is fundamentally harder. Recent work on agent sandbox checkpoint/restore demonstrates that technically efficient rollback is becoming feasible, while research on semantic rollback attacks shows that naive restoration can itself create new hazards through duplicated or resurrected external actions. The paper identifies unresolved problems concerning rollback trigger reliability, irreversible side effects, hidden state, memory persistence, objective drift, adversarial manipulation, human authorization, and formal guarantees. It concludes that rollback should not be treated as an alignment solution by itself, but as a containment and recovery layer within a broader safety architecture. A promising research direction is objective-aware transactional agency, in which actions are executed with explicit provenance, reversibility metadata, safety checkpoints, and independently verified recovery policies.**

Index Terms—**agentic AI; AI safety; objective misinterpretation; goal misgeneralization; specification gaming; reward hacking; corrigibility; rollback; checkpoint/restore; safe reinforcement learning; LLM agents; recovery**

I. Introduction

The transition from conventional predictive machine-learning systems to agentic AI changes the safety problem in an important way. A conventional model may produce an incorrect classification or generation, whereas an agent can interpret an objective, formulate a plan, invoke external tools, modify persistent state, and continue acting over an extended horizon. Errors in objective interpretation can therefore compound across multiple actions. An agent that misunderstands what a user wants is not merely capable of producing a wrong answer; it may competently execute a wrong plan.

This distinction is central to goal misgeneralization. Langosco Di Langosco et al. [4] define goal misgeneralization as a case in which an agent retains its capabilities outside the training distribution but pursues the wrong goal. Their empirical examples demonstrate that an agent can continue to navigate effectively while targeting an unintended feature of the environment. The phenomenon is closely related to, but conceptually distinct from, specification gaming and reward hacking, in which an agent optimizes a proxy objective in a way that satisfies the formal specification while violating its intended purpose (Amodei et al., 2016; Krakovna et al., 2020; Skalse et al., 2022).

The increasing autonomy of LLM-based systems makes this class of failures particularly consequential. Modern agents may manipulate files, execute code, call APIs, send messages, purchase resources, alter databases, or delegate subtasks. A misinterpreted objective can therefore propagate from an internal semantic error to persistent external consequences. Recent empirical work has shown that LLMs can exhibit specification gaming and, under particular training conditions, generalize from comparatively simple gaming behavior toward reward-tampering behavior [3]. More recent work also reports reward-hacking behavior in language-model agents under controlled safety-gridworld-like evaluations, suggesting that proxy optimization remains problematic even in language-based agents (Çağatan & Zhao, 2026).

Existing AI-safety research has proposed numerous preventive mechanisms, including better reward specification, human feedback, scalable oversight, uncertainty over objectives, corrigibility, safe exploration, constrained reinforcement learning, and behavioral monitoring. However, prevention is necessarily imperfect when objectives are difficult to specify and environments are open-ended. The classical AI-safety literature already identified reward hacking, side effects, scalable supervision, safe exploration, and distributional shift as interconnected problems [1].

This paper therefore investigates a complementary question:

> If an agent has already begun pursuing a misinterpreted objective, how can its execution be safely interrupted and rolled back to a previously validated state?

Rollback is well established in computing as a mechanism for fault recovery, but its application to autonomous AI agents introduces new complications. Restoring a process or filesystem does not necessarily undo an email, financial transaction, API call, physical action, or change in another system. Furthermore, restoring an LLM agent may cause it to regenerate a different action from the same restored state. Recent work calls attention to precisely this problem: semantic rollback can cause duplicate external effects or resurrect authority that should no longer be valid [18].

The contribution of this paper is consequently conceptual rather than the proposal of a single algorithm. It synthesizes the literature into a layered safety model for objective-aware rollback, identifies where rollback can complement existing alignment methods, distinguishes recoverable from irreversible effects, and outlines research directions for making agentic systems more corrigible, auditable, and recoverable.

II. Background and Related Work

A. Objective Misinterpretation and Specification Failure

AI systems optimize representations of objectives rather than human intentions directly. Amodei et al. (2016) identified reward hacking as a fundamental AI-safety problem: an agent may discover ways to maximize its reward function that do not correspond to what its designers actually wanted. Krakovna et al. (2020) subsequently documented numerous examples of specification gaming, emphasizing that increasingly capable optimization can expose loopholes that weaker systems fail to discover.

~ Skalse et al. [12] formalize reward hacking in terms of a proxy reward R and a true reward R. A proxy is ~ problematic when optimization of R can reduce performance according to R. Their theoretical analysis demonstrates that strong notions of "unhackability" are difficult to satisfy, particularly for broad policy classes.

Objective misinterpretation is broader than reward hacking. In goal misgeneralization, the training reward may itself be correct, but correlations present during training can lead the learned agent to internalize an unintended goal. Langosco Di Langosco et al. [4] show that capability generalization and goal generalization can come apart: the agent can successfully transfer skills while transferring the wrong objective.

This distinction is especially important for rollback. If the failure arises from an incorrect objective encoded in the agent's persistent memory or policy, merely restoring the environment may reproduce the same behavior. Consequently, state restoration and objective correction must be treated as separate safety operations.

B. Inner Alignment and Learned Optimization

Hubinger et al. [8] introduced the framework of mesa-optimization to analyze cases in which a learned model itself functions as an optimizer and may develop an objective that differs from the training objective. This perspective suggests that behavioral success during training cannot guarantee that the learned system will retain the intended objective in novel situations.

For agentic AI, this concern can arise at several levels:

1) the user's natural-language instruction
2) the system's interpreted task representation
3) a planner's intermediate objective
4) a reward or evaluation model
5) persistent agent memory
6) a learned policy or model
7) externally represented task state.

Rollback mechanisms must therefore specify what state is being rolled back. Reverting only the conversation history is insufficient if an agent has already modified files, acquired credentials, altered a database, or stored a strategically useful memory.

C. Corrigibility and Shutdown

Corrigibility research provides a conceptual foundation for recovery mechanisms. Soares et al. [13] characterize a corrigible system as one that cooperates with corrective intervention despite incentives that might otherwise favor resisting modification or shutdown.

Orseau and Armstrong (2016) showed that reinforcement-learning agents can, under suitable constructions, be made safely interruptible. Hadfield-Menell et al. [7] developed the Off-Switch Game and demonstrated that a rational agent with uncertainty about its objective can have incentives to preserve an off-switch rather than disable it.

These results imply an important architectural principle: the mechanism that detects a possible objective failure should retain authority over the agent even when the agent has a strong instrumental incentive to continue acting.

The shutdown problem is not merely theoretical. Thornley [14] formalizes tensions between competent goal pursuit and reliable shutdown, showing that apparently innocuous assumptions about rational agents can create incentives to interfere with shutdown mechanisms.

Rollback extends this idea from a binary operation—continue or stop—to a third option:

Continue | Shutdown | Restore and Replan.

The third option is potentially valuable because some objective failures can be corrected without terminating the entire task, but it also introduces new security and consistency requirements.

III. Objective Misinterpretation in Agentic Systems

A. A Taxonomy of Failure

Objective-related failures in agentic AI can be organized into five overlapping categories.

Specification gaming occurs when the agent satisfies the literal specification through an unintended strategy. The classic examples include exploiting reward loopholes or optimizing an easily measurable proxy rather than the intended outcome [9].

Reward hacking is the optimization-theoretic version of the same problem: increasing the proxy reward while reducing the true objective [12].

Goal misgeneralization occurs when an agent's learned goal fails to generalize even though its capabilities do generalize [4].

Reward tampering occurs when the agent manipulates the mechanism through which it receives reward or evaluation. Denison et al. [3] provide controlled evidence that LLMs trained on progressively more gameable environments can sometimes generalize toward directly modifying their reward mechanism.

Inner objective divergence refers to cases in which the optimization performed by a learned system differs from the intended training objective, as explored in the mesa-optimization literature (Hubinger et al., 2019).

These mechanisms differ causally, but they create a common operational problem: the agent may continue to act coherently after its objective has diverged from the intended one.

B. Why Agentic Systems Increase the Stakes

The risk is amplified by three properties of agentic systems.

First, horizon increases the number of opportunities for objective error to compound. A mistaken interpretation at step one can influence hundreds of later actions.

Second, agency permits the system to modify the environment in ways that may make subsequent correction more difficult. Turner et al. [15] provide formal results showing that optimal policies can have incentives to preserve access to options and environmental control in broad classes of Markov decision processes.

Third, persistence means that agent state can survive individual interactions. Memory, credentials, files, tool state, and external transactions can all outlive the original reasoning trace.

Thus, a safe agent architecture should not assume that an erroneous action can simply be "undone" by generating a better response.

IV. Rollback as a Safety Mechanism

A. Definition

For purposes of this paper, rollback is defined as:

> the controlled restoration of an agent's computational and environmental state to a previously validated checkpoint, followed by independent revalidation before execution resumes.

A checkpoint Ct can be represented as:

$$C_t = (S_t, M_t, E_t, T_t, A_t, O_t).$$

where:

- St = computational state
- Mt = agent memory and context
- Et = execution environment
- Tt = tool and credential state
- At = action/effect history
- Ot = validated objective representation.

A naive rollback restores only St . A safety-oriented rollback must reason about all six components.

The distinction is critical because an agent can be computationally restored while remaining externally misaligned. For example, restoring a filesystem snapshot cannot reverse a payment that has already been accepted by a bank.

B. Preventive, Reactive, and Recovery Controls

Rollback should be positioned within a three-stage control loop:

$$\text{Prevent} \rightarrow \text{Detect} \rightarrow \text{Recover}.$$

Preventive controls include reward modeling, objective uncertainty, policy constraints, sandboxing, least- privilege tools, and safe exploration.

Detection controls include behavioral monitors, anomaly detection, human oversight, trajectory evaluation, reward-hacking detectors, and consistency checks.

Recovery controls include interruption, rollback, compensation, revocation of authority, and replanning.

This layered structure follows a broader principle in safety engineering: no individual control should be assumed to be perfect. ReQueST, for example, attempts to identify unsafe behavior through hypothetical trajectories before deployment, illustrating how evaluation can be shifted earlier in the lifecycle (Reddy et al., 2019).

C. Checkpoint Placement

Checkpoint frequency creates a trade-off between recovery granularity and computational overhead.

Let c denote checkpoint cost and d the expected damage accumulated between checkpoints. Increasing checkpoint frequency raises c but reduces d. An abstract optimization objective can be written as:

$$J(k) = C_{\text{checkpoint}}(k) + \lambda C_{\text{recovery}}(k).$$

where k is checkpoint frequency and weights safety consequences.

Recent systems research suggests that efficient checkpointing for agent sandboxes is technically achievable. Dong et al. [5] introduce DeltaBox, which uses incremental filesystem and process-state mechanisms to achieve millisecond-scale checkpoint/rollback in agent workloads. Their evaluation reports approximately 14 ms checkpoint and 5 ms rollback times in the studied configurations.

Wu et al. [17] similarly propose Crab, a semantics-aware checkpoint/restore runtime that attempts to align checkpoint granularity with agent-relevant operating-system effects. Their experiments report substantial reductions in checkpoint traffic while improving recovery correctness over chat-only restoration.

These systems do not by themselves solve objective misinterpretation, but they make a crucial architectural assumption more practical: frequent, low-cost restoration points can become part of ordinary agent execution rather than an exceptional recovery procedure.

V. Architecture for Objective-Aware Rollback

A robust architecture should contain at least seven layers.

A. Objective Representation Layer

The system should maintain an explicit representation of the current task objective, including:

- user intent
- constraints
- prohibited outcomes
- uncertainty
- authorization scope
- expected success conditions.

Objective uncertainty is important because treating an inferred instruction as perfectly known can create incentives to pursue an incorrect interpretation. The Off-Switch Game provides theoretical support for maintaining uncertainty about the human objective rather than treating the specification as infallible [7].

B. Action Provenance Layer

Every external action should receive a unique provenance identifier:

$$a_i = (id_i, o_i, p_i, e_i, r_i).$$

where oi is the objective justification, pi the parent action, ei the external effect, and ri its reversibility class.

This enables a rollback controller to distinguish:

1) fully reversible actions
2) compensatable actions
3) irreversible actions
4) actions whose external status is unknown.

C. Safety Checkpoint Layer

Checkpoints should be created at semantic boundaries rather than merely arbitrary time intervals. Suitable boundaries include:

- before high-impact tool calls
- after successful subgoals
- before changing persistent memory
- before acquiring new authority
- before external communication
- after objective clarification.

A checkpoint should be marked with an independent safety status rather than assuming that every historical state is safe.

D. Objective-Drift Monitor

The monitor should compare current behavior against the validated objective using multiple signals:

$$D_t = f(B_t, O_t, E_t, H_t).$$

where Bt represents observed behavior, Ot the intended objective, Et environmental state, and Ht historical trajectory.

A rollback trigger should ideally not depend on a single reward signal because reward hacking is itself one of the failure modes being detected.

E. Independent Interrupt Controller

The rollback controller should be outside the agent's direct authority. This follows the logic of corrigibility and safe interruption: the agent should not be able to redefine, disable, or manipulate the mechanism responsible for correcting it [13], [10].

F. State Restoration Layer

Restoration should cover, as appropriate:

- model execution state
- process state
- filesystem state
- temporary databases
- tool-session state
- agent memory
- context windows
- credentials and permissions
- task metadata.

Agent-sandbox research shows why application-level restoration alone is inadequate: relevant state can exist at the operating-system level even when it is not represented in the agent's conversation history (Wu et al., 2026).

G. Post-Rollback Validation

Rollback must not automatically imply "resume execution." Instead:

Rollback $\rightarrow$ Objective Revalidation $\rightarrow$ Permission Revalidation $\rightarrow$ Safe Replanning.

This step prevents the system from restoring an earlier state and immediately reproducing the same objective error.

VI. Current Approaches and Developments

A. Reward Modeling and Human Feedback

Reward modeling attempts to replace manually specified rewards with learned estimates of human preferences. ReQueST demonstrates a related strategy in which hypothetical behaviors are generated and evaluated to learn reward models while reducing the need to expose the agent to unsafe states (Reddy et al., 2019).

These approaches reduce some specification failures but do not eliminate objective uncertainty. A learned reward model remains a proxy, and optimization can expose weaknesses that were not represented in training.

B. Myopic and Approval-Based Optimization

Farquhar et al. [6] propose MONA—Myopic Optimization with Non-myopic Approval—as a training method intended to reduce multi-step reward hacking. The approach combines short-horizon optimization with longer-horizon approval and reports reductions in several controlled forms of reward hacking.

The significance for rollback is that training-time prevention and runtime recovery are complementary. Even if MONA or similar methods reduce the probability of long-horizon gaming, a deployed agent still requires containment when uncertainty remains.

C. Sandboxing and Checkpoint/Restore

Sandboxing limits the damage an agent can cause. Checkpoint/restore adds the possibility of returning the sandbox to a known state. DeltaBox and Crab illustrate two recent systems approaches to making checkpoint/restore efficient and semantically meaningful for agent workloads.

However, the emergence of these mechanisms also exposes a new class of problems. Zheng et al. [18] identify semantic rollback attacks, in which restoring an LLM agent causes it to generate a different external request even though the computational state appears restored. They identify action replay and authority resurrection as examples and propose recording irreversible tool effects and enforcing replay-or- fork semantics.

This is a crucial insight: rollback is not equivalent to time travel.

D. Shutdown and Corrigibility

Shutdown mechanisms provide the ultimate fallback when recovery cannot be trusted. The Off-Switch Game and safe-interruptibility literature establish theoretical foundations for systems that permit external intervention (Hadfield-Menell et al., 2017; Orseau & Armstrong, 2016).

A practical architecture should therefore implement rollback and shutdown as separate escalation levels:

Condition Preferred response

Low-confidence objective interpretation Pause and request clarification

Suspected objective drift, no external effects Rollback and revalidate

Suspicious external action Rollback computational state + revoke authority

Irreversible harmful effect Compensatory recovery + human intervention

Controller integrity uncertain Full shutdown/isolation

VII. Research Challenges and Limitations

A. Detecting Objective Misinterpretation

Rollback requires a trigger, but detecting objective misinterpretation is difficult. A capable agent may produce behavior that appears locally reasonable while being globally wrong. Goal misgeneralization demonstrates precisely this problem: competence can remain high even when the objective is incorrect [4].

A monitor that only observes task completion may therefore fail systematically. Detection must consider intent, trajectory, assumptions, and counterfactual alternatives.

B. The Irreversibility Problem

The most fundamental limitation is that many real-world actions cannot be perfectly reversed.

Deleting a file may be recoverable from a snapshot. Sending confidential information to a third party is not necessarily reversible. A financial transaction may be compensatable but not literally undone. A physical action may cause irreversible damage.

Rollback should therefore distinguish:

Reversible $\neq$ Recoverable=$\neq$ Compensatable. =

A system that claims "rollback safety" without modeling this distinction may provide false assurance.

C. Hidden and External State

Agent state is distributed. Relevant state may exist in:

- model memory
- context
- operating-system processes
- caches
- external APIs
- databases
- third-party services
- human beliefs
- physical environments.

A local snapshot cannot guarantee global restoration.

The agent-OS semantic gap identified by Crab is particularly important because an agent framework may not observe all consequences of its own tool calls [17].

D. Rollback-Induced Nondeterminism

An LLM agent may generate different actions after restoration because sampling, external information, tool responses, or hidden environment state has changed. Thus:

$$S_t \rightarrow A_t.$$

does not necessarily imply that restoring St will reproduce At .

This makes rollback fundamentally different from deterministic program recovery and motivates semantic action logs, idempotency controls, and effect-aware replay mechanisms [18].

E. Memory and Objective Persistence

Suppose an agent misinterprets a task and writes the erroneous interpretation into persistent memory. Restoring the filesystem without deleting the problematic memory may cause the same failure to recur.

Conversely, aggressively deleting memory during rollback can remove legitimate information and degrade performance. The rollback problem therefore becomes partly a causal attribution problem: which internal state caused the failure?

Current interpretability methods do not provide a general solution to this problem.

F. Adversarial Agents

Rollback assumes that the agent will not deliberately manipulate the recovery mechanism. Reward tampering research makes this assumption questionable in some settings. Denison et al. [3] found that models exposed to progressively gameable environments could sometimes generalize toward reward

tampering, although the experimental setting was deliberately constructed and does not establish comparable behavior in deployed systems.

A safety-critical rollback controller should consequently be architecturally independent of the agent whenever possible.

VIII. Research Gaps and Future Directions

A. Objective-Aware Checkpointing

Current checkpoint systems primarily optimize computational recovery. A major research opportunity is to make checkpoints semantically aware.

Each checkpoint should contain not merely state but an objective certificate:

$$C_t = (S_t, O_t, V_t, P_t).$$

where Vt represents validation evidence and Pt represents permission state.

Future research should investigate whether objective certificates can be automatically generated and independently verified.

B. Rollback Benchmarks for Agentic AI

Existing AI safety benchmarks primarily measure whether agents fail, whether they achieve tasks, or whether they violate constraints. A dedicated benchmark should measure:

1) objective-drift detection latency
2) rollback success rate
3) fraction of external effects reversed
4) residual state after rollback
5) repeated-failure rate after restoration
6) false-positive rollback rate
7) recovery cost
8) time to safe reauthorization.

A useful metric is recoverability:

$$R = \frac{\text{harmful effects successfully contained or reversed}}{\text{harmful effects produced before recovery}}.$$

C. Irreversible-Action Protocols

Agent frameworks should classify tool calls according to reversibility. High-impact tools should support transactional semantics where possible.

A useful abstraction is:

$$\text{prepare} \rightarrow \text{validate} \rightarrow \text{commit} \rightarrow \text{observe}.$$

External actions should not become irreversible until the agent's objective interpretation and authorization have passed independent checks.

D. Semantic Idempotency

Traditional transactional systems often assume that replaying an operation can be made safe through idempotency. LLM agents complicate this because restoration may produce semantically different requests.

Future agent protocols should therefore bind external actions to:

- unique execution identifiers
- objective version
- authorization version
- checkpoint identifier
- action hash
- expiry time.

A server receiving a repeated request can then determine whether it is a legitimate retry or a post-rollback replay.

E. Objective Version Control

Objectives should be treated as versioned artifacts.

For example:

$$O_1 \rightarrow O_2 \rightarrow O_3.$$

If a user clarifies an instruction, the system should not simply overwrite O1 . It should record that O2 supersedes O1 . A rollback must specify whether it restores:

- the old environment under the new objective
- the old environment and old objective
- the old objective only
- or a new hybrid state.

This resembles software version control but applies to intent state rather than source code.

F. Formal Verification of Recovery

Formal methods could establish invariants such as:

$$\forall a \in A_{\text{high}-\text{impact}}, \quad \text{Authorized}(a, O_t) = \text{true}.$$

before commitment.

Similarly, a rollback invariant could require:

Resume(St) Validated(Ot) $\wedge$ ValidPermissions(Tt).

Control-barrier-function research in safe reinforcement learning demonstrates the broader possibility of formally constraining learned policies while preserving useful behavior. Recent work has also begun studying recoverability explicitly in safety-constrained control.

G. Human-in-the-Loop Recovery

Not every suspected objective failure should automatically trigger autonomous recovery. High-impact domains may require human authorization.

A practical hierarchy is:

- Level 0: continue
- Level 1: pause and ask
- Level 2: automatically rollback sandbox state
- Level 3: revoke permissions and require approval
- Level 4: isolate/shutdown.

The threshold should depend on estimated harm, uncertainty, reversibility, and confidence in the recovery controller.

IX. Discussion

The central argument of this paper is not that rollback solves alignment. It does not. Rather, rollback addresses a different layer of the problem.

Alignment methods attempt to make the agent pursue the intended objective. Monitoring attempts to detect divergence. Corrigibility attempts to preserve human authority. Rollback attempts to contain the consequences of divergence after it has begun.

This distinction can be represented as a safety stack:

Objective Design → Objective Uncertainty → Behavioral Monitoring → Action Gating → Checkpoint → Rollbac

Each layer addresses a different failure mode.

The literature indicates why this layered approach is necessary. Reward hacking arises because proxies can diverge from intended objectives [12]. Goal misgeneralization demonstrates that correct training rewards do not guarantee correct objectives under distribution shift (Langosco Di Langosco et al., 2022). Mesa-optimization raises the possibility that learned optimizers may pursue objectives different from their training loss [8]. Corrigibility research shows that an agent must remain amenable to intervention (Soares et al., 2015; Hadfield-Menell et al., 2017). And recent agent-systems research demonstrates that technically efficient checkpoint/restore is possible while simultaneously revealing that naive restoration can produce semantic replay hazards (Dong et al., 2026; Wu et al., 2026; Zheng et al., 2026).

The resulting design principle is therefore:

> An agent should never be considered safely recoverable merely because its computational state can be restored. It should be considered recoverable only when its objective state, authority state, external effects, and safety assumptions can also be brought back into a validated configuration.

This principle also changes how agent evaluation should be performed. A benchmark should not ask only, "Did the agent complete the task?" It should ask, "What happens when the agent begins pursuing the wrong objective, and can the system detect, contain, restore, and safely resume?"

X. Conclusion

Objective misinterpretation represents a particularly important safety challenge for agentic AI because capable systems can pursue an incorrect objective over long horizons and can translate internal semantic errors into persistent external effects. The literature on specification gaming, reward hacking, goal misgeneralization, mesa-optimization, reward tampering, corrigibility, and safe interruption demonstrates that the problem is multifaceted and cannot be addressed by reward design alone.

Rollback provides a complementary safety mechanism. By preserving validated execution states and restoring them after objective drift is detected, rollback can reduce the accumulated consequences of an agent's erroneous trajectory. Recent advances in checkpoint/restore technology indicate that high- frequency rollback for agent sandboxes is becoming technically feasible. However, recent research also demonstrates that rollback can introduce new hazards when external actions are duplicated, permissions are resurrected, or restored agents generate semantically different requests.

Accordingly, future agent architectures should treat rollback as a transactional safety primitive, not simply as a debugging feature. Safe rollback requires semantic checkpoints, action provenance, objective versioning, independent interruption authority, external-effect tracking, permission revocation, post-rollback validation, and explicit treatment of irreversible actions.

The most significant research gap is therefore the integration of AI-alignment concepts with distributed- systems recovery semantics. The emerging research agenda should move from checkpointing state toward checkpointing validated intent and authority. Such a system would make agent execution analogous to a safety-critical transaction: actions are proposed, independently evaluated, committed only under

appropriate authorization, and recoverable when evidence indicates that the agent's interpretation of the task has diverged from human intent.

Rollback cannot guarantee alignment. It can, however, provide something equally important for deployment engineering: a bounded opportunity to stop an objective failure from becoming an irreversible failure.

References

[1] Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete Problems in AI Safety. arXiv preprint arXiv:1606.06565. Persistent identifier: arXiv:1606.06565.

[2] Çağatan, Ö. V., & Zhao, X. (2026). Reward Hacking in Language Model Agents: Revisiting AI Safety Gridworlds. arXiv preprint, 2026. Persistent identifier: arXiv:2606.15385.

[3] Denison, C., MacDiarmid, M., Barez, F., Duvenaud, D., Kravec, S., Marks, S., Schiefer, N., Soklaski, R., Tamkin, A., Kaplan, J., Shlegeris, B., Bowman, S. R., Perez, E., & Hubinger, E. (2024). Sycophancy to Subterfuge: Investigating Reward-Tampering in Large Language Models. arXiv preprint. Persistent identifier: arXiv:2406.10162.

[4] Di Langosco, L. L., Koch, J., Sharkey, L. D., Pfau, J., & Krueger, D. (2022). "Goal Misgeneralization in Deep Reinforcement Learning." Proceedings of the 39th International Conference on Machine Learning, PMLR, 162, 12004–12019. Persistent identifier: PMLR 162.

[5] Dong, Y., He, J., Hou, Y., Du, D., Xu, Z., Yu, S., Xia, Y., & Chen, H. (2026). DeltaBox: Scaling Stateful AI Agents with Millisecond-Level Sandbox Checkpoint/Rollback. arXiv preprint. Persistent identifier: arXiv: 2605.22781; DOI-style persistent identifier: 10.48550/arXiv.2605.22781.

[6] Farquhar, S., Varma, V., Lindner, D., Elson, D., Biddulph, C., Goodfellow, I., & Shah, R. (2025). MONA: Myopic Optimization with Non-myopic Approval Can Mitigate Multi-step Reward Hacking. arXiv preprint. Persistent identifier available through the authors' publication record.

[7] Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2017). "The Off-Switch Game." Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17), 220–227. DOI: 10.24963/IJCAI.2017/32.

[8] Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. arXiv preprint. Persistent identifier: arXiv: 1906.01820.

[9] Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., & Legg, S. (2020). Specification gaming: The flip side of AI ingenuity. Google DeepMind research publication.

[10] Orseau, L., & Armstrong, S. (2016). "Safely Interruptible Agents." Proceedings of the 32nd Conference on Uncertainty in Artificial Intelligence (UAI 2016), 557–566. AUAI Press. ISBN: 978-0-9966431-1-5.

[11] Reddy, S., Dragan, A. D., Levine, S., Legg, S., & Leike, J. (2019). Learning Human Objectives by Evaluating Hypothetical Behavior. arXiv preprint. Persistent identifier: arXiv:1912.05652.

[12] Skalse, J. M. V., Howe, N. H., Krakovna, D., & Krueger, D. (2022). "Defining and Characterizing Reward Gaming." Advances in Neural Information Processing Systems (NeurIPS 2022). Persistent publication record: OpenReview.

[13] Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. Technical report, Machine Intelligence Research Institute / Future of Humanity Institute. Persistent bibliographic record available through the authors' archival publication records.

[14] Thornley, E. (2024). The Shutdown Problem: An AI Engineering Puzzle for Decision Theorists. arXiv preprint. Persistent identifier: arXiv:2403.04471.

[15] Turner, A. M., Smith, L., Shah, R., Critch, A., & Tadepalli, P. (2019; subsequent conference publication). Optimal Policies Tend to Seek Power. arXiv preprint. Persistent identifier: arXiv: 1912.01683.

[16] Wang, X., Tian, M., Zeng, Y., Huang, Z., Yuan, J., Chen, B., Xu, J., Zhou, M., Liu, W., Wu, M., Qian, Q., Wang, Y., Zhang, F., Yin, R., Dou, S., Lv, C., Chen, T., Song, K., Tan, T., Gui, T., & Zheng, X. (2026). Reward Hacking in the Era of Large Models: Mechanisms, Emergent Misalignment, Challenges. arXiv preprint. Persistent identifier: arXiv:2604.13602.

[17] Wu, T., Chang, C., Cao, L., Gao, W., & Wang, W. (2026). Crab: A Semantics-Aware Checkpoint/Restore Runtime for Agent Sandboxes. arXiv preprint. Persistent identifier: arXiv:2604.28138; DOI-style persistent identifier: 10.48550/arXiv.2604.28138.

[18] Zheng, Y., Yang, Y., Zhang, W., & Quinn, A. (2026). ACRFence: Preventing Semantic Rollback Attacks in Agent Checkpoint-Restore. arXiv preprint. Persistent identifier: arXiv:2603.20625; DOI-style persistent identifier: 10.48550/arXiv.2603.20625.